EDUSPARK EXAMS CONSULT

MATHEMATICS
Essay and Objective
2hours

1&2

Name: _____

Index number: _____

Date: _____

B.E.C.E 2027 STEPPING STONE MOCK – SEPTEMBER, 2026

MATHEMATICS [100 marks]

*Write your **name** and **date** in **ink** in the spaces provided above*
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

- ✓ *This booklet consists of two papers, Paper **2** and paper **1***
- ✓ *Answer paper **2** in your answer booklet and paper **1** in the objective test answer sheet*
- ✓ *Paper **2** will last 1hour.*
- ✓ *Paper **1** will last 1hour.*
- ✓ ***Do not start paper 1 until you are told to do so***

EDUSPARK

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PAPER 2
ESSAY
[60 MARKS]

Answer **four** questions **only** from this part
All questions carry equal marks

All working must be clearly shown. Marks will not be awarded for correct answers without corresponding working.

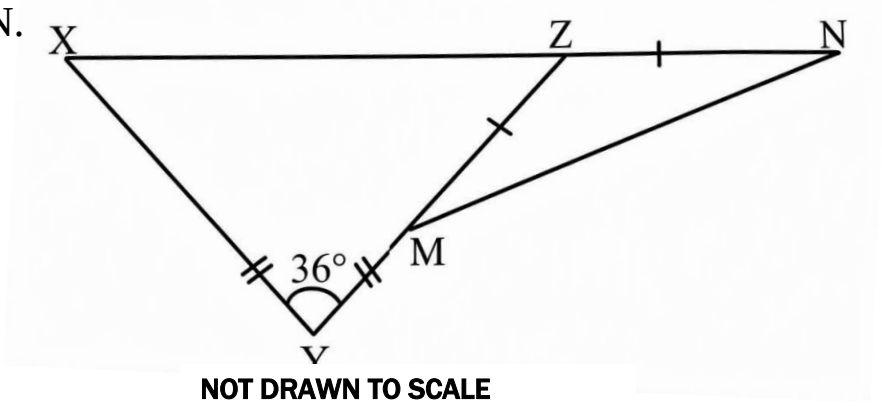
1. a) Given that $9^{2x-1} = \frac{1}{27}$, find the value of x

   b) In how many years will GH¢200.00 invested at 4% per annum simple interest amount to GH¢600.00?

   c) Simplify $3x - (p - x) - (r - p)$

2. a) A rectangular plot of land of width 5km requires a wire mesh of length 36km to fence it. What is the length of the plot?

   b) In the diagram below, |XY| = |YZ|, |MZ| = |NZ| and angle XYZ = 36°. Find the size of angle ZMN.



NOT DRAWN TO SCALE

   c) If A = $\frac{1}{2}$b(a+b), make **a** the subject of the relation

3. a)Find the value of y if $2^y + 1763^0 = 17$

   b) Given the vectors u = $\begin{pmatrix} x+3 \\ 2 \end{pmatrix}$, v = $\begin{pmatrix} y \\ x+y \end{pmatrix}$ and w = $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. If w = u – v, find the values of x and y.

   c) John and Peter are playing a game. Peter has six hundred and nine points. John has thirty – five fewer points than Peter. The winner is the first person to get one thousand points. How many more points does John need to win the game?

4. a) A farmer has 4 bags of seeds **P** = {Maize, Rice, Yam, Cassava} for his trial plot. He wants to try **exactly two** types together. List the sets of all the possible pairs he can try.

   b) In basic 9, learners picked numbers for a game out of $\boldsymbol{\mu} = \{1, 2, 3, \ldots., 14\}$. If the;
   learners who **won a Math prize**, A = {Prime numbers less than 10},
   learners who **won a Sports prize**, B = {Even numbers less than 10} and
   learners who **were present**, C = {x: $3 \leq x < 10, where\ x\ is\ an\ integer$},
   list the elements of ;
   i. learners who **won Math prize** and **were present**
   ii. learners who **won Sports prize** and **were present**
   iii. learners who **were present** but **won No prize**

5. The monthly income of Mr. Yeboah, a senior manager is **GH¢7,200.00**. His monthly expenditure is as follows:
   Fuel - GH¢800.00
   Food - GH¢2,000.00
   Clothing - GH¢700.00
   Wife's Assurance - GH¢1,400.00
   Children's education - GH¢1,600.00
   Savings - GH¢700.00

   a) Represent the information on a pie chart.
   b) This month, his son's school fees increased by GH¢400.00. Mr. Yeboah does not want to touch his Savings and Food. He decides to pay the extra GH¢400.00 by reducing Fuel and Wife's Assurance equally. What will be the new values for Fuel and Wife's Assurance?
   c) What will be the new angles for Fuel and Wife's Assurance on the new pie chart?

6. a) A bakery sells $\frac{2}{3}$ of its daily bread production to a restaurant. If the bakery produces 240 loaves per day, and the restaurant buys bread at GH₵17.50 per loaf, how much money does the bakery receive weekly from the restaurant?

   b) Find the solution set of the inequality $\frac{4x-3}{2} \geq \frac{8x-10}{8} + 2\frac{3}{4}$ if x has a domain {-3, -2, -1, 0, 1, 2, 3, 4, 5}.

   c) A ladder is kept at a distance of 15cm from the wall such that the top of the ladder is at the height of 8cm from the bottom of the wall. Find the length of the ladder.

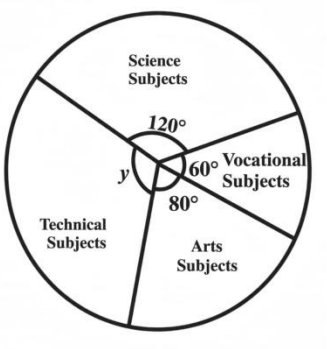
   d) Simplify $6\frac{2}{3} \div (3\frac{4}{15} - 1\frac{3}{5})$

PAPER 1 – OBJECTIVE TEST [30marks]

*Each question is followed by four options lettered **A** to **D**. Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only **one** answer to each question*

The pie chart shows the distribution of programmes offered by 720 students at Kofikrom.

Use this information to answer Questions **1** and **2.**



1. Find the value of the angle marked y.

A. 90°
B. 100°
C. 110°
D. 120°

2. How many more students offered Science subjects than Arts subjects?

A. 40
B. 80
C. 160
D. 240

3. Simplify $2\times(3\frac{1}{3}+1\frac{1}{6})$.

A. $2\frac{1}{3}$
B. $4\frac{1}{2}$
C. $6\frac{2}{3}$
D. 9

4. An amount of GH₵ 375,000.00 was needed to build a clinic for a community of twelve towns. Each community contributed GH₵ 25,000.00. If the District Assembly also contributed GH₵ 30,500.00 , how much more is needed to build the clinic?

A. GH₵ 44,500.00
B. GH₵ 45,500.00
C. GH₵ 34,500.00
D. GH₵ 75,000.00

5. Simplify $\frac{5^7\times 5^4}{5^5}$

A. 5^7
B. 5^8
C. 5^6
D. 5^3

6. Express 0.00348 in standard form.

A. 3.48×10^6
B. 3.48×10^{-3}
C. 3.48×10^{-4}
D. 4.4×10^{-3}

7. Priscilla's age is x years, while Mary's age is 6 years. If Mary is 15 years older than Priscilla, which equation is correct?

A. x + 15= 6
B. x + 6 = 12
C. x - 6 = 15

D. x - 6 = 15

8. Factorize completely: 2xy - 8x + 5y - 20.

A. (2x + 5)(y + 4)
B. (2x - 5)(y + 4)
C. (2x + 5)(y - 4)
D. (2x - 5)(y - 4)

9. If y = c + bx, find y when c = 14/5, b = 3/4, and x = 2.

A. 4
B. 6
C. 7
D. 8

10. Find the gradient of the line joining points A(-3, 5) and B(7, -2).

A. 10
B. -7
C. -7/10
D. -10/7

11. Solve the inequality:

$2x + 10 \geq \frac{3x}{2} - 5$.

A. $x \leq -2$
B. $x \leq -4$
C. $x \geq -30$
D. $x \geq -4$

12. Find m if 4(m + 4) = 8.

A. 6
B. -2
C. 2
D. 4

13. If 480 pupils (80% of enrollment) are boys, find the total number of pupils.

A. 384
B. 540
C. 600
D. 864

14. If 16:8 = r:48, find r.

A. 96
B. 4
C. 8
D. 12

15. Eight men can do a piece of work in 12 days. How long will 6 men take?

A. 14 days
B. 16 days
C. 18 days
D. 20 days

16. An angle greater than 180° but less than 360° is

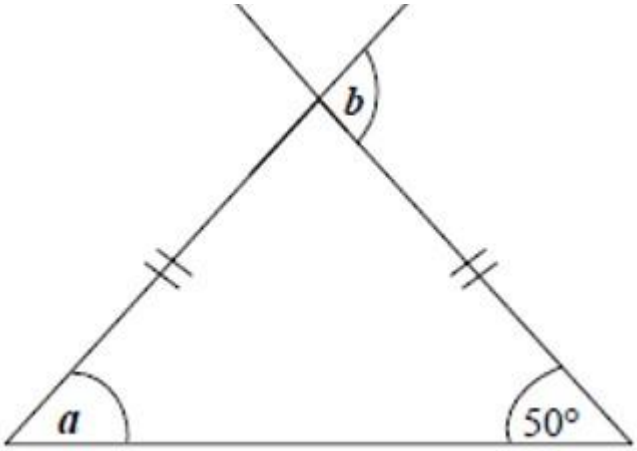
A. Right angle
B. Acute angle
C. Obtuse angle
D. Reflex angle

17. How many edges does a triangular prism have?

A. 3
B. 5
C. 6
D. 9

18. Find the volume of a cube with side 5 cm.

A. 10 cm^3
B. 15 cm^3
C. 25 cm^3
D. 125 cm^3



Use the diagram below to answer Questions 19 and 20

19. Find the angle marked a

A. 70°
B. 50°
C. 40°
D. 30°

20. Find the angle marked b

A. 150°
B. 140°
C. 110°
D. 100°

21. If S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, find the probability that a number selected at random from S is odd.

A 3/8
B 1/4
C 1/2
D 5/8

22. Find the vector which translates the point (4, –5) to (3, –2)

A. $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

B. $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 7 \end{pmatrix}$

D. $\begin{pmatrix} 7 \\ 4 \end{pmatrix}$

23. Factorize completely the expression 2xy-6y+7x-21

A (x-3)(2y+7)
B (x+3)(2y-7)
C (y+3)(2x-7)
D. (y+3)(2x-7)

24. The area of a circle is 154 cm^2. Find the diameter. [Take $\pi = 22/7$]

A 7 cm
B 14 cm
C 21 cm
D 49 cm

25. Maame Esi rides her bicycle to school and back everyday. If the distance from her home to the school is 2345 m, how many kilometers does she cover everyday?

A. 4.98 km
B. 4.69 km
C. 3.96 km
D. 3.68 km

26. The length of a rectangular fence is 25 m. The ratio of the length to the width is 5:3. Find the width of the rectangular fence.

A. 9 m
B. 13 m
C. 15 m
D. 16 m

27. Evaluate $\frac{20}{a} - b$, if a = 30 and b = 1.

A. $-1\frac{2}{3}$
B. $-\frac{1}{3}$
C. $\frac{1}{3}$
D. $1\frac{2}{3}$

28. How many 15Gp Christmas cards can be bought with GH¢18.00?

A. 120
B. 150
C. 180
D. 270

29. If $u = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$ *and* $v = \begin{pmatrix} 4 \\ -5 \end{pmatrix}$, find u+v

$A. u = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

$B. u = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

$C. u = \begin{pmatrix} 10 \\ -14 \end{pmatrix}$

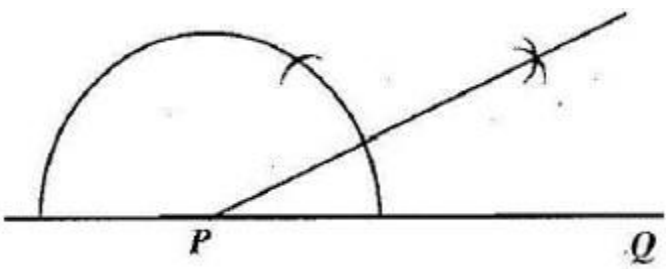
$D. u = \begin{pmatrix} 10 \\ 4 \end{pmatrix}$

30. If 4956 × 25 = 123,900, evaluate 495.6 × 2.5 leaving the answer in standard form.

$A. 1.239 \times 10^2$
$B. 1.239 \times 10^3$
$C. 1.239 \times 10^4$
$D. 1.239 \times 10^5$

31. In an office, 2/3 of the telephone bill is paid by Tom, 1/5 by Azuma and the remaining by Tina. What fraction is paid by Tina?

A 2/15
B 1/4
C 1/3
D 7/15



31. Which of the following best describes the construction?

A. Constructing a perpendicular at P
B. Constructing the bisector of line PQ
C. Constructing an angle of 30° at P
D. Constructing an angle of 45° at P

32. Express 0.055 as a common fraction

A 11/40
B 5/18
C 1/40
D 11/200

The table below shows the distribution of workers in some trades.

Trade	Number of workers
Shoe making	300,000
Mining	25,000
Road transport	160,000
Agriculture	225,000
Manufacturing goods	165,000

Use this information to answer Questions **33** and **34**

33. Which trade employed the most number of workers?

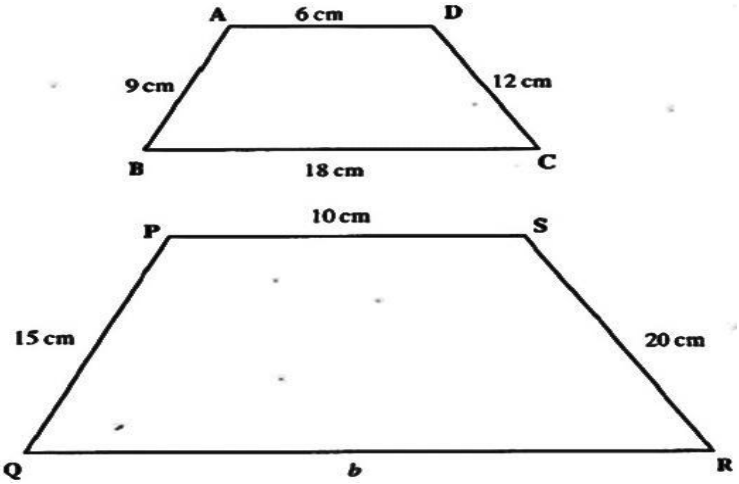
A. Agriculture
B. Manufacturing goods
C. Shoe making
D. Road transport

34. How many people are employed under all the trades?

A. 325,000
B. 485,000
C. 650,000
D. 875,000

35. Aba bought a carton of fish at GH¢ 80.00 and sold it at a profit of GH¢ 13.60. Find the selling price.

A. GH¢ 66.40
B. GH¢ 93.60
C. GH¢ 103.60
D. GH¢ 144.00 37.



36. If the two figures ABCD and PQRS are similar, find the value of b.

A. 60 cm
B. 40 cm
C. 33 cm
D. 30 cm

37. A man shared an amount of money between his two children, Esi and Ato in the ratio 2:3 respectively. If Ato received GH¢ 45.00, what was the total amount shared?

A. GH¢ 18.00
B. GH¢ 27.50
C. GH¢ 75.00
D. GH¢ 112.50

38. How many edges has a cuboid?

A. 16
B. 12
C. 8
D. 4

39. Two sets whose intersection is an empty set are

A. disjoint sets
B. equivalent sets
C. finite sets
D. empty sets

40. If 180 oranges were shared among Kwame and Ama in the ratio 7:5, respectively, how many oranges did Ama receive?

A. 45
B. 60
C. 75
D. 90

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BECE 2027 STEPPING STONE MOCK

MATHEMATICS

FINAL MARKING SCHEME

SECTION A (OBJECTIVE TEST)

1. B	6. B	11. C	16. D	21. C	26. C	31. C	36. D
2. B	7. A	12. B	17. D	22. A	27. A	32. D	37. C
3. D	8. C	13. C	18. D	23. A	28. D	33. C	38. B
4. A	9. A	14. A	19. B	24. B	29. B	34. D	39. A
5. C	10. C	15. B	20. D	25. B	30. A	35. B	40. C

PAPER 2

ESSAY

[60 MARKS]

1. a) Given that $9^{2x-1}=\frac{1}{27}$, find the value of x

$9^{2x-1}=\frac{1}{27}$

$= 3^{2(2x-1)}=\frac{1}{3^3}$ M1

$= 3^{4x-2} = 3^{-3}$ M1

Since the base are the same

4x – 2 = -3 M1

4x = -3 + 2 M1

4x = -1

$x = -\frac{1}{4}$ A1

[5marks]

b) In how many years will GH¢200.00 invested at 4% per annum simple interest amount to GH¢600.00?

The principal, (P) = GH¢200

Rate, I = 4%

Amount, (A) = GH¢600

The simple interest,

$I = \frac{P\times R\times T}{100}$ M1

$I = \frac{200\times 4\times T}{100}$ M1

$100\times I = \frac{200\times 4\times T}{100} \times 100$ M0.5

100I = 800T

$I = \frac{800T}{100} = 8T$ A1

The amount (A) = P + I M1

∴600=200 +8T M0.5

$600 - 200 = 8T$ M0.5

400 = 8T M0.5

$T = \frac{400}{8} = 50$years A1

[7*marks*]

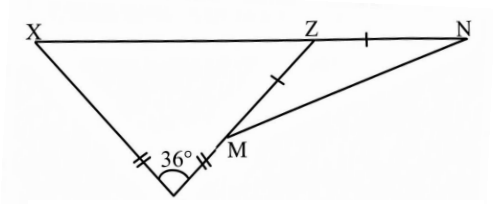
c) Simplify $3x - (p - x) - (r - p)$

$3x - (p - x) - (r - p)$
$= 3x -$ p + x – r +p M1
$= 3x + x - p + p - r$ M1
$= 4x - r$ A1

[$3marks$]

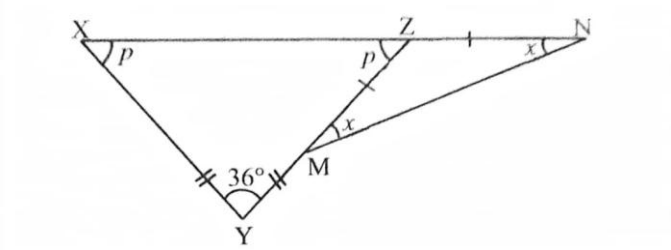
2. a) A rectangular plot of land of width 5km requires a wire mesh of length 36km to fence it. What is the length of the plot?
For a rectangle, the wire mesh needed to fence it = its Perimeter.
Given:
Width, W = 5km
Perimeter, P = 36km
Let Length = L
P = 2(L + W)
36 = 2(L + 5)
36 = 2L + 10
36 – 10 = 2L
26 = 2L
L = $\frac{26}{2}$
L = 13km
The length of the plot is 13 km.

b) In the diagram below, |XY| = |YZ|, |MZ| = |NZ| and angle XYZ = 36°. Find the size of angle ZMN.



NOT DRAWN TO SCALE

Triangles XYZ and MNZ are isosceles.
Let ∠YXZ = ∠YZX = p
Let ∠ZMN = ∠ZNM = x



From triangle XYZ
p + p + 36 = 180 => 2p = 144
p = 72°

Adjacent angles on a straight line add up to 180°
∠MZN = 180° - p = 180° - 72° = 108°

From triangle MNZ
x + x + 108° = 180° => 2x = 72
x = 36°
∠ZMN = 36°

c) If A = $\frac{1}{2}$b(a+b), make **a** the subject of the relation

A = $\frac{1}{2}$b(a+b)

2×A = 2×$\frac{1}{2}$b(a+b)

2A = b(a+b)

2A = ab + b^2

2A - b^2 = ab

a = $\frac{2A-b^2}{b}$

a = $\frac{2A}{b}$ - $\frac{b^2}{b}$

a = $\frac{2A}{b}$ – b

3. a) Find the value of y if $2^y + 1763^0 = 17$

$2^y + 1763^0 = 17$

$2^y + 1 = 17$ M1

$2^y = 17 – 1$ M1

$2^y = 16$ M1

$2^y = 2^4$ M1

$y = 4$ A1

[5$marks$]

b) Given the vectors u = $\begin{pmatrix} x+3 \\ 2 \end{pmatrix}$, v = $\begin{pmatrix} y \\ x+y \end{pmatrix}$ and w = $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. If w = u – v, find the values of x and y.

u = $\begin{pmatrix} x+3 \\ 2 \end{pmatrix}$, **v** = $\begin{pmatrix} y \\ x+y \end{pmatrix}$, **w** = $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

w = u – v

$\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} x+3 \\ 2 \end{pmatrix} - \begin{pmatrix} y \\ x+y \end{pmatrix}$ M0.5

By equality of vectors,

2 = x + 3 – y ……..(1) M0.5

- 1 = 2 – x - y ……(2) M0.5

From (1)

y = x + 3 – 2 M0.5

putting y into (2)

-1 = 2 – x – (x + 3 – 2) M0.5

-1 = 2 – x – x – 3 + 2

-1 = 1 – 2x M0.5

-1 – 1 = -2x M0.5

-2 = -2x M0.5

x = 1 A0.5

putting x into (1)

2 = 1 + 3 – y M0.5

2 – 1- 3 = -y

- 2 = - y M0.5

y = 2 M0.5

[6marks]

c) John and Peter are playing a game. Peter has six hundred and nine points. John has thirty – five fewer points than Peter. The winner is the first person to get one thousand points. How many more points does John need to win the game?

Peter's points = 609

John's points = 609 – 35 = 574 M1

Points needed to win the game = 1000 M1

Points needed by John to win the game = 1000 – 574 = 426 M1

Therefore, John needs 426 points more to win the game. A1

[4marks]

4. a) A farmer has 4 bags of seeds **P** = {Maize, Rice, Yam, Cassava} for his trial plot. He wants to try **exactly two** types together. List the sets of all the possible pairs he can try.
The pairs are:
{Maize, Rice} A0.5
{Maize, Yam} A0.5
{Maize, Cassava} A0.5
{Rice, Yam} A0.5
{Rice, Cassava} A0.5
{Yam, Cassava} A0.5

[3marks]

b) In basic 9, learners picked numbers for a game out of $\boldsymbol{\mu} = \{1, 2, 3, \ldots, 14\}$. If the; learners who **won a Math prize**, A = {Prime numbers less than 10},
learners who **won a Sports prize**, B = {Even numbers less than 10} and
learners who **were present**, C = $\{x: 3 \leq x < 10, where\ x\ is\ an\ integer\}$,
list the elements of ;

i. learners who **won Math prize** and **were present**
ii. learners who **won Sports prize** and **were present**
iii. learners who **were present** but **won No prize**

μ = {1,2,3,4,5,6,7,8,9,10,11,12,13,14} M1
A = {2,3,5,7} M1
B = {2,4,6,8} M1
C = {3,4,5,6,7,8,9} M1

iii. learners who won Math prize and were present = A ∩ C = {3,5,7} A2

ii. learners who won Sports prize and were present = B ∩ C = {4,6,8} A2

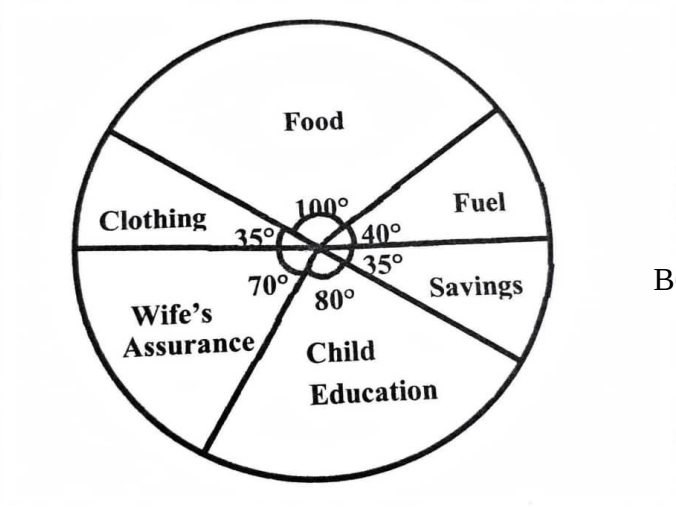
iii. learners who **were present** but **won No prize** = $(A \cup B)' \cap C$ M2
= {1,9,10,11,12,13,14} ∩ {3,4,5,6,7,8,9} M1
= {9} A1

[12marks]

5. The monthly income of Mr. Yeboah, a senior manager is **GH¢7,200.00**. His monthly expenditure is as follows:
Fuel - GH¢800.00
Food - GH¢2,000.00
Clothing - GH¢700.00
Wife's Assurance - GH¢1,400.00
Children's education - GH¢1,600.00
Savings - GH¢700.00
a) Represent the information on a pie chart.

Total Expenditure = GH¢7,200.00

Item	Expenditure	Angle of sector
Fuel	800	$\frac{800}{7200} \times 360 = 40°$ A1
Food	2000	$\frac{2000}{7200} \times 360 = 100°$ A1
Clothing	700	$\frac{700}{7200} \times 360 = 35°$ A1
Wife's Assurance	1400	$\frac{1400}{7200} \times 360 = 70°$ A1
Children's education	1600	$\frac{1600}{7200} \times 360 = 80°$ A1
Savings	700	$\frac{700}{7200} \times 360 = 35°$ A1



B6

b) This month, his son's school fees increased by GH¢400.00. Mr. Yeboah does not want to touch his Savings and Food. He decides to pay the extra GH¢400.00 by reducing Fuel and Wife's Assurance equally. What will be the new values for Fuel and Wife's Assurance?

Extra needed = GH¢400
Reduce Fuel and Wife equally = 400 / 2 = GH¢200 each

New Fuel = 800 - 200 = GH¢600 A1
New Wife's Assurance = 1400 - 200 = GH¢1,200 A1

c) What will be the new angles for Fuel and Wife's Assurance on the new pie chart?
New Fuel angle = $\frac{600}{7200} \times 360 = 30°$ A0.5
New Wife's angle = $\frac{1,200}{7200} \times 360 = 60°$ A0.5

6. a) A bakery sells 2/3 of its daily bread production to a restaurant. If the bakery produces 240 loaves per day, and the restaurant buys bread at GH₵17.50 per loaf, how much money does the bakery receive weekly from the restaurant?

Daily bread production = 240 loaves
Bread sold to restaurant = $\frac{2}{3} \times 240 = 160$ loaves M1
Price per loaf = GH₵17.50
Daily revenue = 160 loaves × GH₵17.50 = GH₵2,800 M1
Weekly revenue (7 days) = GH₵2,800 × 7 M1
= GH₵19,600

So, the bakery receives GH₵19,600 weekly from the restaurant. A1

[4marks]

b) Find the solution set of the inequality $\frac{4x-3}{2} \geq \frac{8x-10}{8} + 2\frac{3}{4}$ if x has a domain {-3, -2, -1, 0, 1, 2, 3, 4, 5}.

$\frac{4x-3}{2} \geq \frac{8x-10}{8} + 2\frac{3}{4}$

$\frac{4x-3}{2} \geq \frac{8x-10}{8} + \frac{11}{4}$ M0.5

$8(\frac{4x-3}{2}) \geq 8\left(\frac{8x-10}{8}\right) + 8 \times \frac{11}{4}$ M0.5

$4(4x - 3) \geq 8x - 10 + 22$ M0.5

$16x - 12 \geq 8x - 10 + 22$ M0.5

$16x - 8x \geq -10 + 22 + 12$ M0.5

$8x \geq 24$ M0.5

$x \geq \frac{24}{8}$

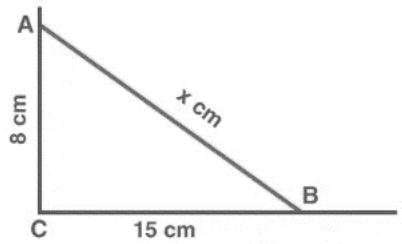
$x \geq 3$

Therefore the solution set for the inequality is {3, 4, 5} A1

[4marks]

c) A ladder is kept at a distance of 15cm from the wall such that the top of the ladder is at the height of 8cm from the bottom of the wall. Find the length of the ladder.

Let |AB| be the ladder of length x cm



B0.5

By Pythagoras theorem

$|AB|^2 = |AC|^2 + |BC|^2$ M0.5

$x^2 = 8^2 + 15^2$ M0.5

$x^2 = 64 + 225$ M0.5

$x^2 = 289$ M0.5

$x = \sqrt{289}$ M0.5

x = 17cm A1

Therefore, the length of the ladder is 17cm

[4marks]

d) Simplify $6\frac{2}{3} \div (3\frac{4}{15} - 1\frac{3}{5})$

$6\frac{2}{3} \div (3\frac{4}{15} - 1\frac{3}{5})$

$= \frac{20}{3} \div (\frac{49}{15} - \frac{8}{5})$ M0.5

$= \frac{20}{3} \div (\frac{49-24}{15})$ M0.5

$= \frac{20}{3} \div \frac{25}{15}$ M0.5

$= \frac{20}{3} \times \frac{15}{25}$ M0.5

$= \frac{20}{5}$ M0.5

= 4 A0.5

[3marks]